

Spatial transcriptomics: foundations and the spot-based workflow

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Day 1 – 16.06.2026

Format each day: about 45 minutes of lecture, a short hands-on briefing, then two hours at the keyboard. Notebooks intentionally hold more than fits in two hours, so they double as a take-home reference.

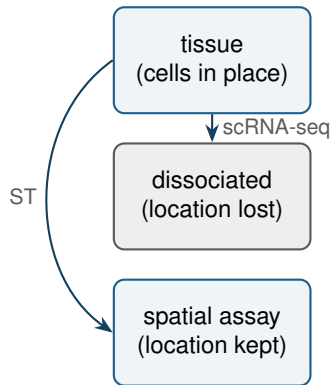
- **Day 1.** The field, the data objects, and a full spot-based (Visium) workflow.
- **Day 2.** Imaging-based single-cell data (Xenium): segmentation, control metrics, and single-cell spatial statistics.
- **Day 3.** Tissue organisation, cell–cell communication, and critical interpretation on a small standalone dataset.

This is a fast-moving field, so we cover the workflow *and* the method landscape, with references to follow up.

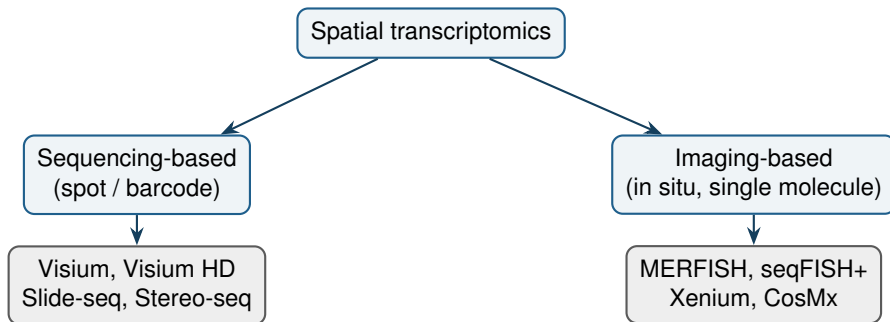
Why spatial?

Dissociation-based scRNA-seq answers *which* cells are present but discards *where* they were.

Tissue function lives in organisation: who sits next to whom, gradients, compartments, and local circuits. Spatial transcriptomics keeps coordinates, so we can ask those questions directly.



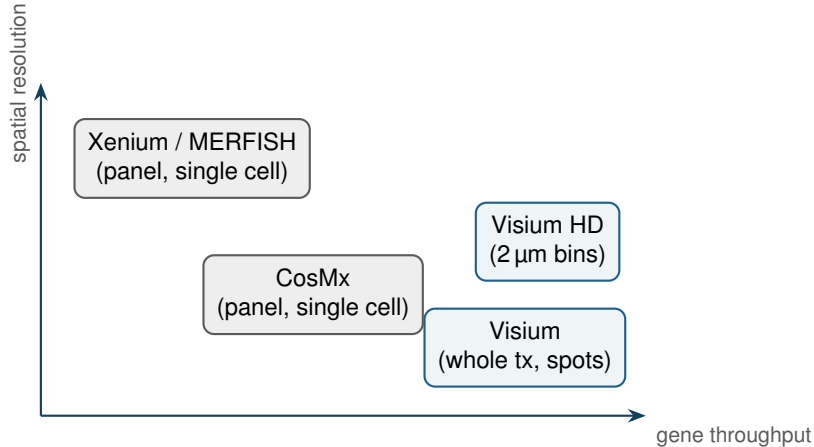
The technology landscape



Two families, two tradeoffs. We use Visium today and Xenium tomorrow as representatives.

Reviews: Marx, *Nat Methods* 2021 (Method of the Year 2020); Moses & Pachter, *Nat Methods* 2022.

The core tradeoff

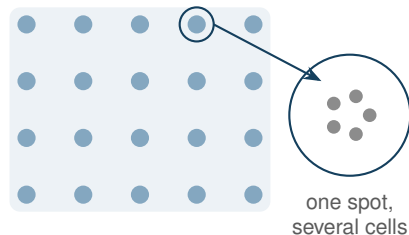


Whole-transcriptome breadth and single-cell resolution still pull against each other.
The right platform depends on the biological question.

How Visium works

- A slide carries a grid of barcoded spots ($\sim 55\mu\text{m}$). Tissue is placed on top, permeabilised, and mRNA is captured under each spot.
- A spot mixes the transcripts of the cells covering it, so a spot is a small *neighbourhood*, not a cell.
- Output: a spot $\times$ gene count matrix, a stained image, and the pixel coordinates of each spot.

Visium HD replaces spots with a lawn of $2\mu\text{m}$ squares, usually binned to $8\mu\text{m}$: near single-cell, still whole-transcriptome.



The data objects

AnnData

X (counts)

obs / var (metadata)

obsm['spatial'] (coords)

uns['spatial'] (image)

obsp (graphs)

AnnData holds one table: counts, per-spot and per-gene metadata, coordinates, image, and graphs. It is enough for one section and is what we use.

SpatialData wraps several AnnData tables with aligned images and shapes in one lazy, larger-than-memory object. It is the right tool for multimodal or multi-sample work, and for Xenium with images.

Wolf et al., *Genome Biol* 2018 (scanpy/AnnData);

Marconato et al., *Nat Methods* 2025 (SpatialData).

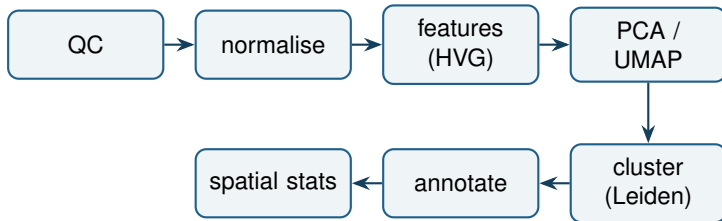
The toolkit (scverse)

- **scanpy** / **anndata**: the single-cell core (QC, normalisation, clustering, plotting).
- **squidpy**: spatial graphs, neighbourhood statistics, image handling.
- **spatialdata** / **spatialdata-io**: readers and a unified object for images, shapes, and tables.

All interoperate, so the spatial workflow extends the familiar single-cell one.

Palla et al., *Nat Methods* 2022 (squidpy); scverse: scverse.org.

The analysis workflow



The first six steps are the standard single-cell pipeline. The spatial layer is added at the end, and revisited once clusters exist.

- Total counts and genes per spot show coverage and flag background spots.
- Mitochondrial fraction marks stressed or low-quality regions.
- Always read each metric *in space*: a bad area is usually a contiguous patch (an edge, a fold), not scattered spots.
- Thresholds are dataset-specific. Look first, then choose, then check what you removed.

[figure slot]

a QC metric (total counts) shown on the tissue, highlighting an edge or fold artefact

suggested source: your own plot from the tutorial, or a QC figure from a methods paper

Normalisation: a landscape, not a default

Method	Idea	When
total-count + log1p	scale to equal depth, log	fast first pass
Pearson residuals	variance-stabilising GLM residuals	sharpen rare states
SCTransform	regularised NB regression	batch/depth confounds

For spots, depth partly reflects how many cells were captured, so normalisation removes some density effect: keep that in mind when reading clusters.

Hafemeister & Satija, *Genome Biol* 2019 (SCTransform); Lause et al., *Genome Biol* 2021 (Pearson residuals).

Dimensionality reduction, clustering, annotation

- PCA then a neighbour graph then UMAP, exactly as in scRNA-seq.
- **Leiden** clustering; resolution sets granularity, and there is no single true value, so choose it for the grain you need.
- Choose the resolution deliberately: scan a range and read a *clustering tree* (clustree / pyclustree). Stable branches are robust populations; clusters that keep splitting and reshuffling signal over-clustering.
- **Annotation**: marker-based naming is transparent and reference-free; label transfer (celltypist, scANVI, ingest) scales when a matched reference exists.

Traag et al., *Sci Rep* 2019 (Leiden); Domínguez Conde et al., *Science* 2022 (CellTypist); Xu et al., *Mol Syst Biol* 2021 (scANVI).

Putting expression in space

- A UMAP shows transcriptional groups; the spatial plot shows whether those groups form coherent tissue regions.
- A cluster that is transcriptionally tight but spatially scattered means something different from one that forms a compartment.

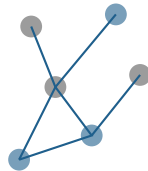
[figure slot]

clusters on the UMAP next to the same clusters on the tissue

suggested source: your own plot from the tutorial

Spatial statistics: first tools

- **Spatial graph:** connect neighbouring spots; this graph underlies every spatial statistic.
- **Neighbourhood enrichment:** which clusters are neighbours more than chance (permutation test). Reveals tissue layering.
- **Spatial autocorrelation:** Moran's I (global smooth trends) and Geary's C (local differences) rank genes that vary across the tissue.



spatial neighbour graph

Spatially variable genes: more methods

- Beyond Moran's I and Geary's C, dedicated models test for spatial structure directly: **SpatialDE** (Gaussian-process), **SPARK** and **SPARK-X** (scalable), and others.
- They differ in null model, speed, and the kind of pattern they favour (smooth gradient vs hotspot).
- For a first pass, Moran's I in squidpy is fast and interpretable.

Moran's I, the simplest test, is

$$I = \frac{N}{W} \frac{\sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}, \quad W = \sum_{ij} w_{ij},$$

with $I > 0$ for smooth spatial trends and null expectation $-1/(N - 1)$.

Svensson et al., *Nat Methods* 2018 (SpatialDE); Sun et al., *Nat Methods* 2020 (SPARK).

Spatial domains: regions, not just clusters

Expression clusters ignore *where* a spot sits. A **spatial domain** is a spatially coherent region (a follicle, a T-cell zone) — coherent in both expression *and* space.

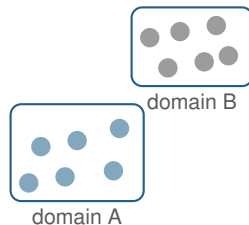
A transparent recipe (nichePCA / BANKSY): average expression over each spot's neighbours and stack it onto its own profile,

$$h_i = [x_i, \lambda \bar{x}_{\mathcal{N}(i)}],$$

then PCA + Leiden. λ trades fine detail for spatial coherence.

Dedicated tools: **BayesSpace**, **SpaGCN**, **STAGATE**, **GraphST**, **BANKSY**.

Zhao et al., *Nat Biotechnol* 2021 (BayesSpace); Hu et al.,



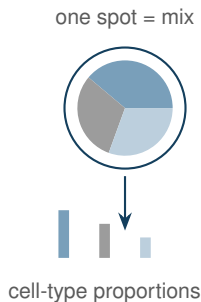
Neighbour smoothing pulls scattered spots into contiguous regions.

The spot problem: deconvolution

A Visium spot mixes several cells, so a “cluster” is a region type, not a cell type. To recover cell-type composition per spot, deconvolution maps a scRNA-seq reference onto the spots.

The baseline is non-negative least squares: with a signature matrix S (genes $\times$ types) and a spot y , solve $\min_{x \geq 0} \|Sx - y\|_2^2$ and normalise x to proportions (we run this in the notebook). Production methods add a count model and platform correction.

Methods: **cell2location**, **RCTD**, **Tangram**, **stereoscope**, **SPOTlight**. Most want a GPU for realistic sizes, so we treat those conceptually and via precomputed results.



What is hard (a preview of the open problems)

- Spot data need deconvolution; imaging data need segmentation. Both inject uncertainty before any biology.
- Targeted panels miss lowly expressed genes (cytokines, chemokines), so immune states are hard to call.
- Spatial QC metrics are still maturing; thresholds are provisional.
- Comparing sections means controlling for composition and batch.

We return to these on Days 2 and 3.

Further reading

- Moses & Pachter. Museum of spatial transcriptomics. *Nat Methods* 2022.
- Marx. Method of the Year 2020: spatially resolved transcriptomics. *Nat Methods* 2021.
- Palla et al. Squidpy. *Nat Methods* 2022.
- Marconato et al. SpatialData. *Nat Methods* 2025.
- Squidpy and SpatialData tutorials: squidpy.readthedocs.io, spatialdata.scverse.org.

How the tutorials work

- Two notebooks per day: `0X_tutorial.ipynb` (worked steps plus blank exercises) and `0X_solutions.ipynb` (filled).
- Worked cells run top to bottom, so run the whole notebook first, then do the exercises, which explore on copies.
- *Method notes* in the notebook frame each choice, so it works as a standalone reference after the course.

Dataset: a public Visium human lymph node section. You will:

- load the section and read the data object;
- compute and inspect QC, then filter;
- normalise, select features, embed, and cluster;
- plot clusters and markers on the tissue;
- run neighbourhood enrichment and Moran's I.

Exercises: choose thresholds, try Pearson-residual normalisation, read a cluster spatially, and compare Geary's C with Moran's I.